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SHARP BOUNDS ON HANKEL DETERMINANTS FOR SAKAGUCHI CLASSES

YONG SUN* AND WEI-PING KUANG

Abstract. We determine the sharp bounds of the second Hankel determinants of logarithmic coefficients for starlike and convex functions with respect to symmetric points, and the sharp bounds of the third Hankel determinants for convex functions with respect to symmetric points.

1. Introduction

Let $\mathcal{A}$ be the class of functions *analytic* in the unit disk $\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$ of the form

$$f(z) = z + \sum_{n=2}^{\infty} a_n z^n. \tag{1.1}$$

We denote $\mathcal{S}$ by the subclass of $\mathcal{A}$ consisting of univalent functions.

Sakaguchi [40] introduced a class $\mathcal{S}_s^*$ of starlike functions with respect to symmetric points, it consists of functions $f \in \mathcal{S}$ satisfying

$$\Re\Big(\frac{2zf'(z)}{f(z)-f(-z)}\Big) > 0 \qquad (z \in \mathbb{D}). \tag{1.2}$$

In a later paper, Das and Singh [13] discussed a class $\mathcal{K}_s$ of convex functions with respect to symmetric points, it consists of functions $f \in \mathcal{S}$ satisfying

$$\Re\Big(\frac{2\big(zf'(z)\big)'}{\big(f(z)-f(-z)\big)'}\Big) > 0 \qquad (z \in \mathbb{D}). \tag{1.3}$$

Given $q, n \in \mathbb{N}$, the Hankel determinant $H_{q,n}(f)$ of $f \in \mathcal{A}$ of the form (1.1) is defined by

$$H_{q,n}(f) = \begin{vmatrix} a_n & a_{n+1} & \cdots & a_{n+q-1} \\ a_{n+1} & a_{n+2} & \cdots & a_{n+q} \\ \vdots & \vdots & \vdots & \vdots \\ a_{n+q-1} & a_{n+q} & \cdots & a_{n+2(q-1)} \end{vmatrix}.$$

In recent years, many papers have been devoted to finding bounds of determinants, whose elements are coefficients of functions in $\mathcal{A}$, or its subclasses. The sharp bounds on the second Hankel determinants $|H_{2,1}(f)|$ and $|H_{2,2}(f)|$ were obtained by [6, 9, 16, 17, 29, 35], for various classes of analytic functions.

Note that for $f \in \mathcal{A}$, $a_1 = 1$, $H_{3,1}(f)$ reduces to

$$H_{3,1}(f) = a_3\big(a_2a_4 - a_3^2\big) - a_4\big(a_4 - a_2a_3\big) + a_5\big(a_3 - a_2^2\big). \tag{1.4}$$

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Obviously, the case of the upper bounds on $|H_{3,1}(f)|$ is much more difficult than the cases of $|H_{2,1}(f)|$ and $|H_{2,2}(f)|$. We refer to [4, 7, 10, 37, 39, 41, 45] for discussions on the upper bounds of the third Hankel determinants $|H_{3,1}(f)|$ for various classes of univalent functions. However, these results are far from sharpness. In a recent paper, Kwon *et al.* [26] found such a formula of expressing c_4 for Carathéodory functions, the sharp results of the third Hankel determinants are found for some classes of univalent functions (see e.g., [5, 21, 22, 23, 27, 28, 38, 43, 44]).

Recently, Krishna *et al.* [24] (see also [8, 36]) obtained the upper bounds of $|H_{3,1}(f)|$ for the classes of starlike and convex functions with respect to symmetric points.

Theorem A. Let $f(z)$ be given by (1.1).

(1) If $f \in \mathcal{S}_s^*$, then $|H_{3,1}(f)| \leq \frac{5}{2}$;

(2) If $f \in \mathcal{K}_s$, then $|H_{3,1}(f)| \leq \frac{19}{135} \approx 0.141$.

Very recently, Cho *et al.* [12] improved the upper bounds of $|H_{3,1}(f)|$ for the class of starlike functions with respect to symmetric points.

Theorem B. If $f \in \mathcal{S}_s^*$ be given by (1.1), then

$$|H_{3,1}(f)| \leq \frac{1}{144}\left(3879 - 2218\sqrt{3} - 1356\sqrt{15+4\sqrt{3}} + 783\sqrt{45+12\sqrt{3}}\right) \approx 0.26547.$$

In this paper, we prove that $|H_{3,1}(f)| \leq 4/135$ for $f \in \mathcal{K}_s$, and so giving the sharp bound for $|H_{3,1}(f)|$ for a significant subclass of $\mathcal{K}_s$.

For $f \in \mathcal{S}$, let

$$F_f(z) := \log\left(\frac{f(z)}{z}\right) = 2\sum_{n=1}^{\infty} \gamma_n(f) z^n \qquad (z \in \mathbb{D};\ \log 1 := 0). \tag{1.5}$$

The numbers $\gamma_n := \gamma_n(f)$ are called logarithmic coefficients of f. It is well known that the logarithmic coefficients play a crucial role in Milin conjecture [32]. Sharp logarithmic coefficient estimates for the class $\mathcal{S}$ are already known for $n = 1$ and $n = 2$, given by $|\gamma_1| \leq 1$ and $|\gamma_2| \leq 1/2 + 1/e^2$, respectively. However, the bound of γ_n for $n \geq 3$, is still an open problem. We refer to [1, 2, 11, 15, 25, 42] for discussions on the logarithmic coefficient for various classes of univalent functions.

Given $q, n \in \mathbb{N}$, the Hankel determinant $H_{q,n}(F_f/2)$ which entries are logarithmic coefficients of $f \in \mathcal{A}$ of the form (1.1) is defined by

$$H_{q,n}(F_f/2) = \begin{vmatrix} \gamma_n & \gamma_{n+1} & \cdots & \gamma_{n+q-1} \\ \gamma_{n+1} & \gamma_{n+2} & \cdots & \gamma_{n+q} \\ \vdots & \vdots & \vdots & \vdots \\ \gamma_{n+q-1} & \gamma_{n+q} & \cdots & \gamma_{n+2(q-1)} \end{vmatrix}.$$

A study of Hankel determinant with entries as logarithmic coefficients was initiated by Kowalczyk and Lecko [18]. Due to the great importance of logarithmic coefficients, the proposed topic seems reasonable and interesting.

By differentiating (1.5) and using (1.1) we get

$$\begin{aligned} \gamma_1 &= \frac{1}{2}a_2, & \gamma_2 &= \frac{1}{2}\left(a_3 - \frac{1}{2}a_2^2\right), \\ \gamma_3 &= \frac{1}{2}\left(a_4 - a_2a_3 + \frac{1}{3}a_2^3\right), & \gamma_4 &= \frac{1}{2}\left(a_5 - a_2a_4 + a_2^2a_3 - \frac{1}{2}a_3^2 - \frac{1}{4}a_2^4\right). \end{aligned} \tag{1.6}$$

Therefore,

$$H_{2,1}(F_f/2) = \gamma_1\gamma_3 - \gamma_2^2 = \frac{1}{4}\left(a_2a_4 - a_3^2 + \frac{1}{12}a_2^4\right). \tag{1.7}$$

and

$$\begin{aligned} &H_{2,2}(F_f/2) = \gamma_2\gamma_4 - \gamma_3^2 \\ =&\frac{1}{4}\left[\left(a_3 - \frac{1}{2}a_2^2\right)a_5 + \left(a_2a_3 - a_4 - \frac{1}{6}a_2^3\right)a_4 + \left(\frac{1}{4}a_2^2a_3 - \frac{1}{2}a_3^2 - \frac{1}{12}a_2^4\right)a_3 + \frac{1}{72}a_2^6\right]. \end{aligned} \tag{1.8}$$

By observing that when $f \in \mathcal{S}$, for $f_\theta(z) := e^{-i\theta}f(e^{i\theta}z)$ with $\theta \in \mathbb{R}$, we have

$$H_{2,2}(F_{f_\theta}/2) = e^{6i\theta}H_{2,2}(F_f/2).$$

Thus, the functional $|H_{2,2}(F_f/2)| = |\gamma_2\gamma_4 - \gamma_3^2|$ is a rotationally invariant.

Recently, Kowalczyk and Lecko [18] obtained the sharp bounds of $|H_{2,1}(F_f/2)|$ for the classes of starlike and convex functions. Moreover, the sharp bounds of $|H_{2,1}(F_f/2)|$ for the classes of starlike and convex functions of order $\alpha(0 \le \alpha < 1)$ were found in [19]. Very recently, Kowalczyk and Lecko [20] examined the sharp bounds of $|H_{2,1}(F_f/2)|$ for the classes of strongly starlike and strongly convex functions. Allu *et al.* [3] (see also [34]) examined the sharp bounds of $|H_{2,1}(F_f/2)|$ for the classes of starlike and convex functions with respect to symmetric points. Moreover, the sharp bounds of $|H_{2,1}(F_f/2)|$ for the class of starlike functions of order $\alpha\,(0 \le \alpha < 1)$ with respect to symmetric points were investigated in [33]. Very recently, Eker *et al.* [14] obtained the sharp bounds of $|H_{2,1}(F_f/2)|$ and $|H_{2,1}(F_{f^{-1}}/2)|$ for the classes of strongly Ozaki close-to-convex functions and inverse functions, respectively.

The problem of finding sharp bounds of $|H_{2,2}(F_f/2)|$ is technically much more difficult. The purpose of this paper is to prove the sharp bounds of $|H_{2,2}(F_f/2)|$ for the classes $\mathcal{S}_s^*$ and $\mathcal{K}_s$, respectively.

Denote $\mathcal{P}$ by the class of *Carathéodory functions* p normalized by

$$p(z) = 1 + \sum_{n=1}^{\infty} c_n z^n \qquad (z \in \mathbb{D}) \tag{1.9}$$

and satisfy the condition $\Re\big(p(z)\big) > 0$.

The following results are known for functions belonging to the class $\mathcal{P}$, which will be required in the proof of our main results.

Lemma 1.1. (See[26, 30, 31]) *If $p \in \mathcal{P}$ and is given by* (1.9) *with $c_1 \ge 0$, then*

$$c_1 = 2\zeta_1, \tag{1.10}$$

$$c_2 = 2\zeta_1^2 + 2(1-\zeta_1^2)\zeta_2, \tag{1.11}$$

$$c_3 = 2\zeta_1^3 + 4(1-\zeta_1^2)\zeta_1\zeta_2 - 2(1-\zeta_1^2)\zeta_1\zeta_2^2 + 2(1-\zeta_1^2)(1-|\zeta_2|^2)\zeta_3 \tag{1.12}$$

and

$$\begin{aligned} c_4 =&2\zeta_1^4 + 2(1-\zeta_1^2)(3\zeta_1^2 + \zeta_2 - 3\zeta_1^2\zeta_2 + \zeta_1^2\zeta_2^2)\zeta_2 \\ &+ 2(1-\zeta_1^2)(1-|\zeta_2|^2)\big(2\zeta_1 - 2\zeta_1\zeta_2 - \overline{\zeta_2}\zeta_3\big)\zeta_3 \\ &+ 2(1-\zeta_1^2)(1-|\zeta_2|^2)(1-|\zeta_3|^2)\zeta_4, \end{aligned} \tag{1.13}$$

for some $\zeta_1 \in [0,\ 1]$ and $\zeta_2, \zeta_3, \zeta_4 \in \overline{\mathbb{D}} := \{z \in \mathbb{C} :\ |z| \le 1\}$.

2. Main results

In this section, we will prove the sharp bounds on the second Hankel determinants of logarithmic coefficients for the Sakaguchi classes $\mathcal{S}_s^*$ and $\mathcal{K}_s$, and the sharp bounds on the third Hankel determinants for the class $\mathcal{K}_s$, respectively.

We begin by deriving the sharp bounds of $|H_{2,2}(F_f/2)|$ for the class $\mathcal{S}_s^*$.

Theorem 2.1. *If $f \in \mathcal{S}_s^*$ be of the form (1.1), then*

$$\big|H_{2,2}(F_f/2)\big| \leq \frac{1}{8}. \tag{2.1}$$

The bound is sharp.

Proof. For the function $f \in \mathcal{S}_s^*$ given by (1.1), there exists an analytic function $p \in \mathcal{P}$ in the unit disk $\mathbb{D}$ with $p(0) = 1$ and $\Re\big(p(z)\big) > 0$ such that

$$\frac{2zf'(z)}{f(z) - f(-z)} = p(z) \qquad (z \in \mathbb{D}). \tag{2.2}$$

By elementary calculations, we have

$$z + \sum_{n=2}^{\infty} n a_n z^n = \left(1 + \sum_{n=1}^{\infty} c_n z^n\right)\left(z + \sum_{n=2}^{\infty} a_{2n-1} z^{2n-1}\right). \tag{2.3}$$

It follows from (2.3) that

$$\begin{aligned} a_2 &= \frac{1}{2}c_1, & a_3 &= \frac{1}{2}c_2, \\ a_4 &= \frac{1}{8}\big(2c_3 + c_1c_2\big), & a_5 &= \frac{1}{8}\big(2c_4 + c_2^2\big). \end{aligned} \tag{2.4}$$

Thus, (1.8) and (2.4) give

$$18432 \cdot H_{2,2}(F_f/2) = 144c_4\big(4c_2 - c_1^2\big) - 288c_3^2 - 24c_1^3c_3 + 72c_1^2c_2^2 - 24c_1^4c_2 + c_1^6.$$

Since the class $\mathcal{S}_s^*$ and $|H_{2,2}(F_f/2)|$ are rotationally invariant, we may assume that $c_1 \in [0, 2]$. Thus, in view of (1.10) we assume that $\zeta_1 \in [0, 1]$. By making use of (1.10)-(1.13), it follows that

$$\begin{aligned} 18432 \cdot H_{2,2}(F_f/2) =& \big[64\zeta_1^6 + 1920(1-\zeta_1^2)\zeta_1^4\zeta_2 + 384(1-\zeta_1^2)(12 - 11\zeta_1^2)\zeta_1^2\zeta_2^2 \\ &+ 1152(1-\zeta_1^2)(2 - 4\zeta_1^2 + 3\zeta_1^4)\zeta_2^3 + 1152(1-\zeta_1^2)^2\zeta_1^2\zeta_2^4\big] \\ &- 384(1-\zeta_1^2)(1-|\zeta_2|^2)\big[(1+6\zeta_2)\zeta_1^3 + 6(1-\zeta_1^2)\zeta_1\zeta_2^2\big]\zeta_3 \\ &- 1152(1-\zeta_1^2)(1-|\zeta_2|^2)\big[(1-\zeta_1^2)(1+|\zeta_2|^2) + \zeta_1^2\overline{\zeta_2}\big]\zeta_3^2 \\ &+ 1152(1-\zeta_1^2)(1-|\zeta_2|^2)(1-|\zeta_3|^2)\big[\zeta_1^2 + 2(1-\zeta_1^2)\zeta_2\big]\zeta_4, \end{aligned}$$

for some $\zeta_1 \in [0, 1]$ and $\zeta_2, \zeta_3, \zeta_4 \in \overline{\mathbb{D}}$. Since $|\zeta_4| \leq 1$, we have

$$\begin{aligned} &288 \cdot \big|H_{2,2}(F_f/2)\big| \\ \leq& \big|\zeta_1^6 + 30(1-\zeta_1^2)\zeta_1^4\zeta_2 + 6(1-\zeta_1^2)(12 - 11\zeta_1^2)\zeta_1^2\zeta_2^2 \\ &+ 18(1-\zeta_1^2)(2 - 4\zeta_1^2 + 3\zeta_1^4)\zeta_2^3 + 18(1-\zeta_1^2)^2\zeta_1^2\zeta_2^4\big| \\ &+ 6(1-\zeta_1^2)(1-|\zeta_2|^2)\big|(1+6\zeta_2)\zeta_1^3 + 6(1-\zeta_1^2)\zeta_1\zeta_2^2\big| \cdot |\zeta_3| \\ &+ 18(1-\zeta_1^2)(1-|\zeta_2|^2)\big[\big|(1-\zeta_1^2)(1+|\zeta_2|^2) + \zeta_1^2\overline{\zeta_2}\big| - \big|\zeta_1^2 + 2(1-\zeta_1^2)\zeta_2\big|\big] \cdot |\zeta_3|^2 \\ &+ 18(1-\zeta_1^2)(1-|\zeta_2|^2)\big|\zeta_1^2 + 2(1-\zeta_1^2)\zeta_2\big|. \end{aligned}$$

A. Suppose that

$$\left|(1-\zeta_1^2)(1+|\zeta_2|^2)+\zeta_1^2\overline{\zeta_2}\right|-\left|\zeta_1^2+2(1-\zeta_1^2)\zeta_2\right|\geq 0.$$

Then

$$288\cdot\left|H_{2,2}(F_f/2)\right|\leq u(\zeta_1,|\zeta_2|),$$

where $u:\ \mathbb{R}^2\to\mathbb{R}$ is defined by

$$\begin{aligned}u(x,y):=&x^6+6(1-x^2)(3-3x^2+x^3)\\&+6(1-x^2)(3x^2+6x^3+5x^4)y\\&+6(1-x^2)(6x+12x^2-7x^3-11x^4)y^2\\&+6(1-x^2)(6-15x^2-6x^3+9x^4)y^3\\&+18(1-x^2)^2(x^2-2x-1)y^4.\end{aligned}$$

We show that $u(x,y)\leq 36$ for $(x,y)\in[0,1]\times[0,1]$.

I. On the vertices of $[0,1]\times[0,1]$, we have

$$u(0,0)=18,\quad u(0,1)=36,\quad u(1,0)=u(1,1)=1.$$

II. On the sides of $[0,1]\times[0,1]$, we get

$$\begin{aligned}&u(0,y)=18(1+2y^3-y^4)\leq u(0,1)=36\quad(y\in(0,1)),\\&u(x,0)=x^6+6x^3(1-x^2)+18(1-x^2)^2\leq u(0,0)=18\quad(x\in(0,1)),\\&u(1,y)=1\quad(y\in(0,1)),\\&u(x,1)=x^6+18(1-x^2)(2+x^2)\leq u(0,1)=36\quad(x\in(0,1)).\end{aligned}$$

III. It remains to consider the set $(0,1)\times(0,1)$. By a numerical computation, all the real solutions ($x\neq 0,\pm1$) of the system of equations

$$\begin{aligned}\frac{\partial u}{\partial x}=&6\big[(-12x+3x^2+10x^3-5x^4+x^5)\\&+(6x+18x^2+8x^3-30x^4-30x^5)y\\&+(6+24x-39x^2-92x^3+35x^4+66x^5)y^2\\&+(-42x-18x^2+96x^3+30x^4-54x^5)y^3\\&+6(1-x^2)(-1+3x+5x^2-3x^3)y^4\big]=0,\end{aligned}$$

and

$$\begin{aligned}\frac{\partial u}{\partial y}=&6(1-x^2)\big[(3x^2+6x^3+5x^4)+2(6x+12x^2-7x^3-11x^4)y\\&+3(6-15x^2-6x^3+9x^4)y^2+12(1-x^2)(x^2-2x-1)y^3\big]=0\end{aligned}$$

can be presented as follows

$$\begin{cases}x_1\approx 2.04563,\\y_1\approx 1.37733,\end{cases}\quad\begin{cases}x_2\approx -0.705084,\\y_2\approx -1.2058,\end{cases}\quad\begin{cases}x_3\approx -1.28019,\\y_3\approx 2.16531,\end{cases}$$

$$\begin{cases}x_4\approx 0.780864,\\y_4\approx -1.21112,\end{cases}\quad\begin{cases}x_5\approx 0.913224,\\y_5\approx -0.766575.\end{cases}$$

Thus, the function u has no critical point in $(0,1)\times(0,1)$.

B. Suppose that

$$\left|(1-\zeta_1^2)(1+|\zeta_2|^2)+\zeta_1^2\overline{\zeta_2}\right|-\left|\zeta_1^2+2(1-\zeta_1^2)\zeta_2\right|<0.$$

Then

$$288 \cdot \left|H_{2,2}(F_f/2)\right| \le v(\zeta_1, |\zeta_2|),$$

where $v:\ \mathbb{R}^2 \to \mathbb{R}$ is defined by

$$\begin{aligned} v(x,y) :=& x^6 + 6(1-x^2)(3x^2+x^3) \\ &+ 6(1-x^2)(6-6x^2+6x^3+5x^4)y \\ &+ 6(1-x^2)(6x+9x^2-7x^3-11x^4)y^2 \\ &+ 6(1-x^2)(-6x^2-6x^3+9x^4)y^3 \\ &+ 18(1-x^2)^2(x^2-2x)y^4. \end{aligned}$$

We show that $v(x,y) \le 36$ for $(x,y) \in [0,1]\times[0,1]$.

I. On the vertices of $[0,1]\times[0,1]$, we have

$$v(0,0) = 0, \quad v(0,1) = 36, \quad v(1,0) = v(1,1) = 1.$$

II. On the sides of $[0,1]\times[0,1]$, we get

$$\begin{aligned} &v(0,y) = 36y \le v(0,1) = 36 \quad (y \in (0,1)), \\ &v(x,0) = x^6 + 6(1-x^2)(3x^2+x^3) \le 36 \quad (x \in (0,1)), \\ &v(1,y) = 1 \quad (y \in (0,1)), \\ &v(x,1) = x^6 + 18(1-x^2)(2+x^2) \le v(0,1) = 36 \quad (x \in (0,1)). \end{aligned}$$

III. It remains to consider the set $(0,1)\times(0,1)$. By a numerical computation, all the real solutions ($x \ne 0, \pm 1$) of the system of equations

$$\begin{aligned} \frac{\partial v}{\partial x} =& 6\big[(6x+3x^2-12x^3-5x^4+x^5) \\ &+ (-24x+18x^2+44x^3-30x^4-30x^5)y \\ &+ (6+18x-39x^2-80x^3+35x^4+66x^5)y^2 \\ &+ (-12x-18x^2+60x^3+30x^4-54x^5)y^3 \\ &+ 6(1-x^2)(-1+x+5x^2-3x^3)y^4\big] = 0, \end{aligned}$$

and

$$\begin{aligned} \frac{\partial v}{\partial y} =& 6(1-x^2)\big[(6-6x^2+6x^3+5x^4) + 2(6x+9x^2-7x^3-11x^4)y \\ &+ 3(-6x^2-6x^3+9x^4)y^2 + 12(1-x^2)(x^2-2x)y^3\big] = 0 \end{aligned}$$

can be presented as follows

$$\begin{cases} x_1 \approx 2.04056, \\ y_1 \approx 1.31635, \end{cases} \quad \begin{cases} x_2 \approx -0.763424, \\ y_2 \approx -0.832764, \end{cases} \quad \begin{cases} x_3 \approx -2.11443, \\ y_3 \approx 0.124877, \end{cases}$$

$$\begin{cases} x_4 \approx -1.33921, \\ y_4 \approx 2.03819, \end{cases} \quad \begin{cases} x_5 \approx -1.84814, \\ y_5 \approx 0.0543994, \end{cases} \quad \begin{cases} x_6 \approx -2.17911, \\ y_6 \approx 0.415374, \end{cases}$$

$$\begin{cases} x_7 \approx -1.40556, \\ y_7 \approx 0.367493, \end{cases} \quad \begin{cases} x_8 \approx 0.9206, \\ y_8 \approx -1.10586, \end{cases} \quad \begin{cases} x_9 \approx 0.724979, \\ y_9 \approx -0.635281. \end{cases}$$

Thus, the function v has no critical point in $(0,1)\times(0,1)$.

Summarizing, we see that the bounds obtained in Parts A and B give

$$\left|H_{2,2}(F_f/2)\right| \leq \frac{1}{288}\cdot 36 = \frac{1}{8}.$$

We finally note that equality in (2.1) holds for the function $f \in \mathcal{S}_s^*$ defined by

$$\frac{2zf'(z)}{f(z)-f(-z)} = \frac{1+z^2}{1-z^2} \quad (z \in \mathbb{D}),$$

for which $a_2 = a_4 = 0$, and $a_3 = a_5 = 1$. The proof is thus completed.

□

Example 2.1. Let f_1 be given by (1.1), and satisfying (2.2) with

$$p(z) := \frac{1+z}{1-z} \qquad (z \in \mathbb{D}),$$

for which $a_2 = a_3 = a_4 = a_5 = 1$. In view of (1.2) and (2.2), we see that $f_1 \in \mathcal{S}_s^*$, and satisfy

$$\left|H_{2,2}(F_{f_1}/2)\right| = \frac{1}{288} < \frac{1}{8}.$$

Next, we consider the sharp bounds of $|H_{2,2}(F_f/2)|$ for the functions $f \in \mathcal{K}_s$.

Theorem 2.2. *If $f \in \mathcal{K}_s$ be of the form* (1.1)*, then*

$$\left|H_{2,2}(F_f/2)\right| \leq \frac{13}{1080}. \tag{2.5}$$

The bound is sharp.

Proof. Let the function $f \in \mathcal{K}_s$ given by (1.1). According to the Alexander relationship

$$f \in \mathcal{K}_s \Longleftrightarrow zf' \in \mathcal{S}_s^*,$$

by applying the formula (2.4), we have

$$\begin{aligned} a_2 &= \frac{1}{4}c_1, & a_3 &= \frac{1}{6}c_2, \\ a_4 &= \frac{1}{32}\big(2c_3 + c_1c_2\big), & a_5 &= \frac{1}{40}\big(2c_4 + c_2^2\big). \end{aligned} \tag{2.6}$$

Thus, (1.8) and (2.6) give

$$\begin{aligned} 17694720\cdot H_{2,2}(F_f/2) =& 2304\big(16c_2 - 3c_1^2\big)c_4 - 17280c_3^2 - 720c_1^3c_3 - 5760c_1c_2c_3 \\ & - 96c_1^2c_2^2 - 1080c_1^4c_2 + 8192c_2^3 + 15c_1^6. \end{aligned}$$

Since the class $\mathcal{K}_s$ and $\left|H_{2,2}(F_f/2)\right|$ are rotationally invariant, we may assume that $c_1 \in [0,2]$. Thus, in view of (1.10) we assume that $\zeta_1 \in [0,1]$. By making use of (1.10)-(1.13), it follows that

$$\begin{aligned} &276480\cdot H_{2,2}(F_f/2) \\ =&\big[-65\zeta_1^6 + 2268(1-\zeta_1^2)\zeta_1^4\zeta_2 + 20(1-\zeta_1^2)(282-273\zeta_1^2)\zeta_1^2\zeta_2^2 \\ &+ 16(1-\zeta_1^2)(208-389\zeta_1^2+271\zeta_1^4)\zeta_2^3 + 1224(1-\zeta_1^2)^2\zeta_1^2\zeta_2^4\big] \\ &- 36(1-\zeta_1^2)(1-|\zeta_2|^2)\big[(5+80\zeta_2)\zeta_1^3 + (1-\zeta_1^2)(12+68\zeta_2)\zeta_1\zeta_2\big]\zeta_3 \\ &- 72(1-\zeta_1^2)(1-|\zeta_2|^2)\big[20\zeta_1^2\overline{\zeta_2} + (1-\zeta_1^2)(15+17|\zeta_2|^2)\big]\zeta_3^2 \\ &+ 288(1-\zeta_1^2)(1-|\zeta_2|^2)(1-|\zeta_3|^2)\big[5\zeta_1^2 + 8(1-\zeta_1^2)\zeta_2\big]\zeta_4, \end{aligned}$$

for some $\zeta_1 \in [0,1]$ and $\zeta_2, \zeta_3, \zeta_4 \in \overline{\mathbb{D}}$. Since $|\zeta_4| \le 1$, we have

$$\begin{aligned}
&276480 \cdot \big|H_{2,2}(F_f/2)\big| \\
\le&\big| -65\zeta_1^6 + 2268(1-\zeta_1^2)\zeta_1^4\zeta_2 + 20(1-\zeta_1^2)(282-273\zeta_1^2)\zeta_1^2\zeta_2^2 \\
&+ 16(1-\zeta_1^2)(208-389\zeta_1^2+271\zeta_1^4)\zeta_2^3 + 1224(1-\zeta_1^2)^2\zeta_1^2\zeta_2^4\big| \\
&+ 36(1-\zeta_1^2)(1-|\zeta_2|^2)\big|(5+80\zeta_2)\zeta_1^3 + (1-\zeta_1^2)(12+68\zeta_2)\zeta_1\zeta_2\big| \cdot |\zeta_3| \\
&+ 72(1-\zeta_1^2)(1-|\zeta_2|^2)\big[\big|20\zeta_1^2\overline{\zeta_2} + (1-\zeta_1^2)(15+17|\zeta_2|^2)\big| - 4\big|5\zeta_1^2 + 8(1-\zeta_1^2)\zeta_2\big|\big] \cdot |\zeta_3|^2 \\
&+ 288(1-\zeta_1^2)(1-|\zeta_2|^2)\big|5\zeta_1^2 + 8(1-\zeta_1^2)\zeta_2\big|.
\end{aligned}$$

A. Suppose that

$$\big|20\zeta_1^2\overline{\zeta_2} + (1-\zeta_1^2)(15+17|\zeta_2|^2)\big| - 4\big|5\zeta_1^2 + 8(1-\zeta_1^2)\zeta_2\big| \ge 0.$$

Then

$$276480 \cdot \big|H_{2,2}(F_f/2)\big| \le \phi(\zeta_1, |\zeta_2|),$$

where $\phi:\ \mathbb{R}^2 \to \mathbb{R}$ is defined by

$$\begin{aligned}
\phi(x,y) :=& 65x^6 + 180(1-x^2)(6-6x^2+x^3) \\
&+ 36(1-x^2)(12x+40x^2+68x^3+63x^4)y \\
&+ 12(1-x^2)(12+204x+458x^2-219x^3-455x^4)y^2 \\
&+ 16(1-x^2)(208-27x-479x^2-153x^3+271x^4)y^3 \\
&+ 1224(1-x^2)^2(x^2-2x-1)y^4.
\end{aligned}$$

We show that $\phi(x,y) \le 3328$ for $(x,y) \in [0,1]\times[0,1]$.
I. On the vertices of $[0,1]\times[0,1]$, we have

$$\phi(0,0) = 1080, \quad \phi(0,1) = 3328, \quad \phi(1,0) = \phi(1,1) = 65.$$

II. On the sides of $[0,1]\times[0,1]$, we get

$$\begin{aligned}
&\phi(0,y) = 3328y^3 + 72(1-y^2)(15+17y^2) \le \phi(0,1) = 3328 \quad (y \in (0,1)), \\
&\phi(x,0) = 65x^6 + 180(1-x^2)(6-6x^2+x^3) \le \phi(0,0) = 1080 \quad (x \in (0,1)), \\
&\phi(1,y) = 65 \quad (y \in (0,1)), \\
&\phi(x,1) = 65x^6 + 16(1-x^2)(208+40x^2-5x^4) \le \phi(0,1) = 3328 \quad (x \in (0,1)).
\end{aligned}$$

III. It remains to consider the set $(0,1)\times(0,1)$. By a numerical computation, all the real solutions ($x \ne 0,\ \pm 1$) of the system of equations

$$\begin{aligned}
\frac{\partial\phi}{\partial x} =& 30(-144x+18x^2+144x^3-30x^4+13x^5) \\
&+ 72(6+40x+84x^2+46x^3-170x^4-189x^5)y \\
&+ 12(204+892x-1269x^2-3652x^3+1095x^4+2730x^5)y^2 \\
&+ 16(-27-1374x-378x^2+3000x^3+765x^4-1626x^5)y^3 \\
&+ 2448(-1+3x+6x^2-6x^3-5x^4+3x^5)y^4 = 0,
\end{aligned}$$

and

$$\begin{aligned}\frac{\partial\phi}{\partial y} =&12(1-x^2)\big[3(12x+40x^2+68x^3+63x^4)\\ &+2(12+204x+458x^2-219x^3-455x^4)y\\ &+4(208-27x-479x^2-153x^3+271x^4)y^2\\ &+408(1-x^2)(x^2-2x-1)y^3\big]=0\end{aligned}$$

can be presented as follows

$$\begin{cases}x_1\approx 1.62865,\\ y_1\approx 1.56547,\end{cases}\quad \begin{cases}x_2\approx -0.752544,\\ y_2\approx -1.97446,\end{cases}\quad \begin{cases}x_3\approx -2.83754,\\ y_3\approx 0.494378,\end{cases}$$

$$\begin{cases}x_4\approx -1.42328,\\ y_4\approx 1.70275,\end{cases}\quad \begin{cases}x_5\approx 0.744909,\\ y_5\approx -1.06728,\end{cases}\quad \begin{cases}x_6\approx -1.98728,\\ y_6\approx 1.15862,\end{cases}$$

$$\begin{cases}x_7\approx 0.880789,\\ y_7\approx -0.721075,\end{cases}\quad \begin{cases}x_8\approx -0.00248494,\\ y_8\approx -0.0306462.\end{cases}$$

Thus, the function ϕ has no critical point in $(0,1)\times(0,1)$.

B. Suppose that

$$\big|20\zeta_1^2\overline{\zeta_2}+(1-\zeta_1^2)(15+17|\zeta_2|^2)\big|-4\big|5\zeta_1^2+8(1-\zeta_1^2)\zeta_2\big|<0.$$

Then

$$276480\cdot\big|H_{2,2}(F_f/2)\big|\le\psi(\zeta_1,|\zeta_2|),$$

where $\psi:\ \mathbb{R}^2\to\mathbb{R}$ is defined by

$$\begin{aligned}\psi(x,y):=&65x^6+180(1-x^2)(8x^2+x^3)\\ &+36(1-x^2)(64+12x-64x^2+68x^3+63x^4)y\\ &+4(1-x^2)(612x+1050x^2-657x^3-1365x^4)y^2\\ &+16(1-x^2)(64-27x-245x^2-153x^3+271x^4)y^3\\ &+1224(1-x^2)^2(x^2-2x)y^4.\end{aligned}$$

We show that $\psi(x,y)\le 3328$ for $(x,y)\in[0,1]\times[0,1]$.

I. On the vertices of $[0,1]\times[0,1]$, we have

$$\psi(0,0)=0,\quad \psi(0,1)=3328,\quad \psi(1,0)=\psi(1,1)=65.$$

II. On the sides of $[0,1]\times[0,1]$, we get

$$\begin{aligned}&\psi(0,y)=2304y+1024y^3\le\psi(0,1)=3328\quad(y\in(0,1)),\\ &\psi(x,0)=65x^6+180x^2(1-x^2)(8+x)\le 3328\quad(x\in(0,1)),\\ &\psi(1,y)=65\quad(y\in(0,1)),\\ &\psi(x,1)=65x^6+16(1-x^2)(208+40x^2-5x^4)\le\psi(0,1)=3328\quad(x\in(0,1)).\end{aligned}$$

III. It remains to consider the set $(0,1)\times(0,1)$. By a numerical computation, all the real solutions ($x\neq 0,\ \pm 1$) of the system of equations

$$\begin{aligned}\frac{\partial\psi}{\partial x} =&30(96x+18x^2-192x^3-30x^4+13x^5)\\ &+36(12-256x+168x^2+508x^3-340x^4-378x^5)y\\ &+4(612+2100x-3807x^2-9660x^3+3285x^4+8190x^5)y^2\\ &+16(-27-618x-378x^2+2064x^3+765x^4-1626x^5)y^3\\ &+2448(-1+x+6x^2-4x^3-5x^4+3x^5)y^4=0,\end{aligned}$$

and

$$\begin{aligned}\frac{\partial\psi}{\partial y} =&(1-x^2)\big[36(64+12x-64x^2+68x^3+63x^4)\\ &+8(612x+1050x^2-657x^3-1365x^4)y\\ &+48(64-27x-245x^2-153x^3+271x^4)y^2\\ &+4896(1-x^2)(x^2-2x)y^3\big]=0\end{aligned}$$

can be presented as follows

$$\begin{cases}x_1\approx -5.95844,\\ y_1\approx 0.277983,\end{cases}\quad \begin{cases}x_2\approx 3.18653,\\ y_2\approx 1.24009,\end{cases}\quad \begin{cases}x_3\approx -0.759642,\\ y_3\approx -1.20212,\end{cases}$$

$$\begin{cases}x_4\approx 0.904134,\\ y_4\approx -1.07128,\end{cases}\quad \begin{cases}x_5\approx 0.755277,\\ y_5\approx -0.738039.\end{cases}$$

Thus, the function ψ has no critical point in $(0,1)\times(0,1)$.

Summarizing, we see that the bounds obtained in Parts A and B give

$$\big|H_{2,2}(F_f/2)\big|\leq \frac{1}{276480}\cdot 3328=\frac{13}{1080}.$$

We finally note that equality in (2.5) holds for the function $f\in\mathcal{K}_s$ defined by

$$\frac{2\big(zf'(z)\big)'}{\big(f(z)-f(-z)\big)'}=\frac{1+z^2}{1-z^2}\quad (z\in\mathbb{D}),$$

for which $a_2=a_4=0$, $a_3=1/3$ and $a_5=1/5$. The proof is thus completed.

□

We finally prove the sharp bounds of $|H_{3,1}(f)|$ for functions $f\in\mathcal{K}_s$.

Theorem 2.3. *If $f\in\mathcal{K}_s$ be of the form* (1.1)*, then*

$$\big|H_{3,1}(f)\big|\leq\frac{4}{135}. \tag{2.7}$$

The bound is sharp.

Proof. Let the function $f\in\mathcal{K}_s$ given by (1.1). Thus, (1.4) and (2.6) give

$$138240\cdot H_{3,1}(f)=144c_4\big(8c_2-3c_1^2\big)-540c_3^2+180c_1c_2c_3-64c_2^3+9c_1^2c_2^2.$$

Since the class $\mathcal{K}_s$ and $\big|H_{3,1}(f)\big|$ are rotationally invariant, we may assume that $c_1\in[0,2]$. Thus, in view of (1.10) we assume that $\zeta_1\in[0,1]$. From (1.10)-(1.13), it follows

that

$$\begin{aligned}
&8640\cdot H_{3,1}(f)\\
=&\big[4\zeta_1^6+156(1-\zeta_1^2)\zeta_1^4\zeta_2+3(1-\zeta_1^2)(163-151\zeta_1^2)\zeta_1^2\zeta_2^2\\
&+2(1-\zeta_1^2)(128-319\zeta_1^2+227\zeta_1^4)\zeta_2^3+153(1-\zeta_1^2)^2\zeta_1^2\zeta_2^4\big]\\
&-2(1-\zeta_1^2)(1-|\zeta_2|^2)\big[18(1+4\zeta_2)\zeta_1^3-(1-\zeta_1^2)(63-153\zeta_2)\zeta_1\zeta_2\big]\zeta_3\\
&-9(1-\zeta_1^2)(1-|\zeta_2|^2)\big[(1-\zeta_1^2)(15+17|\zeta_2|^2)+8\zeta_1^2\overline{\zeta_2}\big]\zeta_3^2\\
&+72(1-\zeta_1^2)(1-|\zeta_2|^2)(1-|\zeta_3|^2)\big[\zeta_1^2+4(1-\zeta_1^2)\zeta_2\big]\zeta_4,
\end{aligned}$$

for some $\zeta_1\in[0,1]$ and $\zeta_2,\zeta_3,\zeta_4\in\overline{\mathbb{D}}$. Since $|\zeta_4|\le 1$, we have

$$\begin{aligned}
&8640\cdot\big|H_{3,1}(f)\big|\\
\le&\big|4\zeta_1^6+156(1-\zeta_1^2)\zeta_1^4\zeta_2+3(1-\zeta_1^2)(163-151\zeta_1^2)\zeta_1^2\zeta_2^2\\
&+2(1-\zeta_1^2)(128-319\zeta_1^2+227\zeta_1^4)\zeta_2^3+153(1-\zeta_1^2)^2\zeta_1^2\zeta_2^4\big|\\
&+2(1-\zeta_1^2)(1-|\zeta_2|^2)\big|18(1+4\zeta_2)\zeta_1^3-(1-\zeta_1^2)(63-153\zeta_2)\zeta_1\zeta_2\big|\cdot|\zeta_3|\\
&+9(1-\zeta_1^2)(1-|\zeta_2|^2)\big[\big|(1-\zeta_1^2)(15+17|\zeta_2|^2)+8\zeta_1^2\overline{\zeta_2}\big|-8\big|\zeta_1^2+4(1-\zeta_1^2)\zeta_2\big|\big]\cdot|\zeta_3|^2\\
&+72(1-\zeta_1^2)(1-|\zeta_2|^2)\big|\zeta_1^2+4(1-\zeta_1^2)\zeta_2\big|.
\end{aligned}$$

A. Suppose that

$$\big|(1-\zeta_1^2)(15+17|\zeta_2|^2)+8\zeta_1^2\overline{\zeta_2}\big|-8\big|\zeta_1^2+4(1-\zeta_1^2)\zeta_2\big|\ge 0.$$

Then

$$8640\cdot\big|H_{3,1}(f)\big|\le h(\zeta_1,|\zeta_2|),$$

where $h:\ \mathbb{R}^2\to\mathbb{R}$ is defined by

$$\begin{aligned}
h(x,y):=&4x^6+9(1-x^2)\big(15-15x^2+4x^3\big)\\
&+6(1-x^2)\big(21x+12x^2+3x^3+26x^4\big)y\\
&+3(1-x^2)\big(6+102x+157x^2-114x^3-151x^4\big)y^2\\
&+2(1-x^2)\big(128-63x-355x^2-9x^3+227x^4\big)y^3\\
&+153(1-x^2)^2\big(x^2-2x-1\big)y^4.
\end{aligned}$$

We show that $h(x,y)\le 256$ for $(x,y)\in[0,1]\times[0,1]$.
I. On the vertices of $[0,1]\times[0,1]$, we have

$$h(0,0)=135,\quad h(0,1)=256,\quad h(1,0)=h(1,1)=4.$$

II. On the sides of $[0,1]\times[0,1]$, we get

$$\begin{aligned}
&h(0,y)=256y^3+9(1-y^2)(15+17y^2)\le h(0,1)=256\quad(y\in(0,1)),\\
&h(x,0)=4x^6+9(1-x^2)\big(15-15x^2+4x^3\big)\le h(0,0)=135\quad(x\in(0,1)),\\
&h(1,y)=4\quad(y\in(0,1)),\\
&h(x,1)=4x^6+4(1-x^2)(x^4+x^2+64)\le h(0,1)=256\quad(x\in(0,1)).
\end{aligned}$$

III. It remains to consider the set $(0,1)\times(0,1)$. By a numerical computation, all the real solutions ($x\neq 0,\ \pm 1$) of the system of equations

$$\begin{aligned}\frac{\partial h}{\partial x} =&\big(-540x+108x^2+540x^3-180x^4+24x^5\big)\\ &+\big(126+144x-324x^2+336x^3-90x^4-936x^5\big)y\\ &+3\big(102+302x-648x^2-1232x^3+570x^4+906x^5\big)y^2\\ &+2\big(-63-966x+162x^2+2328x^3+45x^4-1362x^5\big)y^3\\ &+306\big(-1+3x+6x^2-6x^3-5x^4+3x^5\big)y^4=0,\end{aligned}$$

and

$$\begin{aligned}\frac{\partial h}{\partial y} =&6(1-x^2)\big[\big(21x+12x^2+3x^3+26x^4\big)\\ &+\big(6+102x+157x^2-114x^3-151x^4\big)y\\ &+\big(128-63x-355x^2-9x^3+227x^4\big)y^2\\ &+102(1-x^2)\big(x^2-2x-1\big)y^3\big]=0\end{aligned}$$

can be presented as follows

$$\begin{cases}x_1\approx 4.09147,\\ y_1\approx 3.6864,\end{cases}\quad \begin{cases}x_2\approx 1.54374,\\ y_2\approx 0.697197,\end{cases}\quad \begin{cases}x_3\approx -0.669378,\\ y_3\approx -0.944821,\end{cases}$$

$$\begin{cases}x_4\approx 0.822505,\\ y_4\approx -1.39553,\end{cases}\quad \begin{cases}x_5\approx -1.1847,\\ y_5\approx 1.57982,\end{cases}\quad \begin{cases}x_6\approx 0.927148,\\ y_6\approx -0.713145,\end{cases}$$

$$\begin{cases}x_7\approx -0.012557,\\ y_7\approx -0.0649226,\end{cases}\quad \begin{cases}x_8\approx -2943.38,\\ y_8\approx 0.9703.\end{cases}$$

Thus, the function h has no critical point in $(0,1)\times(0,1)$.

B. Suppose that

$$\big|(1-\zeta_1^2)(15+17|\zeta_2|^2)+8\zeta_1^2\overline{\zeta_2}\big|-8\big|\zeta_1^2+4(1-\zeta_1^2)\zeta_2\big|<0.$$

Then

$$8640\cdot\big|H_{3,1}(f)\big|\leq g(\zeta_1,|\zeta_2|),$$

where $g:\ \mathbb{R}^2\to\mathbb{R}$ is defined by

$$\begin{aligned}g(x,y) :=&4x^6+36x^2(1-x^2)(2+x)\\ &+6(1-x^2)(48+21x-48x^2+3x^3+26x^4)y\\ &+3(1-x^2)(102x+139x^2-114x^3-151x^4)y^2\\ &+2(1-x^2)(-16-63x-175x^2-9x^3+227x^4)y^3\\ &+153(1-x^2)^2(x^2-2x)y^4.\end{aligned}$$

We show that $g(x,y)\leq 256$ for $(x,y)\in[0,1]\times[0,1]$.

I. On the vertices of $[0,1]\times[0,1]$, we have

$$g(0,0)=0,\quad g(0,1)=256,\quad g(1,0)=g(1,1)=4.$$

II. On the sides of $[0,1]\times[0,1]$, we get

$$\begin{aligned}
&g(0,y)=288y-32y^3\le g(0,1)=256 \quad (y\in(0,1)),\\
&g(x,0)=4x^6+36x^2(1-x^2)(2+x)\le 256 \quad (x\in(0,1)),\\
&g(1,y)=4 \quad (y\in(0,1)),\\
&g(x,1)=4x^6+4(1-x^2)(x^4+x^2+64)\le g(0,1)=256 \quad (x\in(0,1)).
\end{aligned}$$

III. It remains to consider the set $(0,1)\times(0,1)$. By a numerical computation, all the real solutions ($x\neq 0,\ \pm1$) of the system of equations

$$\begin{aligned}
\frac{\partial g}{\partial x}=&12\big(12x+9x^2-24x^3-15x^4+2x^5\big)\\
&+6\big(21-192x-54x^2+296x^3-15x^4-156x^5\big)y\\
&+6\big(51+139x-324x^2-580x^3+285x^4+453x^5\big)y^2\\
&+6\big(-21-106x+54x^2+536x^3+15x^4-454x^5\big)y^3\\
&+306\big(-1+x+6x^2-4x^3-5x^4+3x^5\big)y^4=0,
\end{aligned}$$

and

$$\begin{aligned}
\frac{\partial g}{\partial y}=&6(1-x^2)\big[\big(48+21x-48x^2+3x^3+26x^4\big)\\
&+\big(102x+139x^2-114x^3-151x^4\big)y\\
&+\big(-16-63x-175x^2-9x^3+227x^4\big)y^2\\
&+102(1-x^2)\big(x^2-2x\big)y^3\big]=0
\end{aligned}$$

can be presented as follows:

$$\begin{aligned}
&\begin{cases} x_1\approx 3.49847,\\ y_1\approx 4.23066,\end{cases}
&&\begin{cases} x_2\approx 0.925095,\\ y_2\approx -4.46575,\end{cases}
&&\begin{cases} x_3\approx 0.78935,\\ y_3\approx -2.82606,\end{cases}\\
&\begin{cases} x_4\approx -0.841837,\\ y_4\approx -0.622089,\end{cases}
&&\begin{cases} x_5\approx 1.57658,\\ y_5\approx 0.783429,\end{cases}
&&\begin{cases} x_6\approx -1.21911,\\ y_6\approx 1.48873,\end{cases}\\
&\begin{cases} x_7\approx 0.940547,\\ y_7\approx -0.933221,\end{cases}
&&\begin{cases} x_8\approx 0.846863,\\ y_8\approx -0.591323,\end{cases}
&&\begin{cases} x_9\approx -2941.5,\\ y_9\approx 0.971849.\end{cases}
\end{aligned}$$

Thus, the function g has no critical point in $(0,1)\times(0,1)$.

Summarizing, we see that the bounds obtained in Parts A and B give

$$\big|H_{3,1}(f)\big|\le\frac{1}{8640}\cdot 256=\frac{4}{135}.$$

We finally note that equality in (2.7) holds for the function $f\in\mathcal{K}_s$ defined by

$$\frac{2\big(zf'(z)\big)'}{\big(f(z)-f(-z)\big)'}=\frac{1+z^2}{1-z^2}\quad(z\in\mathbb{D}),$$

for which $a_2=a_4=0$, $a_3=1/3$ and $a_5=1/5$. The proof is thus completed.

□

Remark 2.1. Making use of the similar arguments as in Theorem 2.3, we obtain the inequality $|H_{3,1}(f)|\le\alpha\approx 0.27589$ for $f\in\mathcal{S}_s^*$, obviously, the upper bound of $|H_{3,1}(f)|$ in Theorem B is the better one.

Example 2.2. Let f_2 be given by (1.1), and satisfying

$$\frac{2\big(zf_2'(z)\big)'}{\big(f_2(z)-f_2(-z)\big)'}=\frac{1+z^3}{1-z^3}\quad(z\in\mathbb{D}),$$

for which $a_2=a_3=a_5=0$ and $a_4=1/8$. In view of (1.3), we see that $f_2\in\mathcal{K}_s$, and satisfy

$$\big|H_{2,2}(F_{f_2}/2)\big|=\frac{1}{256}<\frac{13}{1080},\qquad \big|H_{3,1}(f_2)\big|=\frac{1}{64}<\frac{4}{135},$$

respectively.

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Conflicts of Interest. The authors declare that they have no conflict of interest.

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